

Notes: Verhoogen (QJE, 2008)

Trade, Quality Upgrading, and Wage Inequality in the Mexican Manufacturing Sector

Contents

1	Big picture	3
2	Incremental contribution	3
3	Data and measurement	4
3.1	Core datasets	4
3.2	Main variables	4
4	Model and theoretical mechanism	4
4.1	Demand and willingness to pay for quality	4
4.2	Production, worker quality, and wages	5
4.3	Optimal plant choices	5
4.4	Export cutoffs and the peso crisis	5
5	Empirical specifications	6
5.1	Cross-sectional regressions	6
5.2	Differential-change regressions	6
5.3	Alternative productivity proxies and robustness	6
6	Identification and empirical design	6
6.1	Core identification logic	6
6.2	Main threats and responses	7
7	Section-by-section study guide	7
8	Tables and figures	7
8.1	Figure I: Wage Inequality, 1984–2001 + Figure II: Wage Variance, EIA 1984–2001 Panel . . .	7
8.2	Figure III: Real Exchange Rate, 1984–2002 + Figure IV: Exports, High-quality Models as Percentage of VW Output	8
8.3	Figure V: Shift toward Exporting, 1993–2001 + Figure VI: Average Quality	9
8.4	Figure VII: Response to Exchange-Rate Devaluation + Figure VIII: Change in Average Quality in Response to Devaluation	9
8.5	Table I: Summary Statistics by Export Status + Table II: Baseline Estimates, EIA 1993–2001 Panel	10
8.6	Table III: Estimates from EIA 1984–2001 Panel + Table IV: Estimates from EIA-ENESTyC Panel	11
8.7	Table V: Estimates Using Alternative Proxies for λ , EIA 1993–2001 Panel + Table VI: Estimates for Cost of Capital, Publicly Listed Mexican Firms, 1989–2000	12

8.8 Table VII: Comparing Non-maquiladora and Maquiladora Sectors	12
9 Result map	13
10 Short summary	14
11 Conclusion	14
11.1 Core conclusion	14
11.2 Economic intuition	14
11.3 Incremental contribution relative to standard trade theory	15
11.4 Evidence assessment	15
11.5 Limitations	15
11.6 Takeaway	15

1 Big picture

- **Question.** Why did wage inequality rise in Mexico during a period of trade liberalization and export expansion, even though a simple Heckscher–Ohlin logic would predict a fall in skill premia for an unskilled-labor-abundant country?
- **Core mechanism.** The paper proposes a **quality-upgrading mechanism**: more productive plants respond to export opportunities by producing higher-quality goods, and higher-quality production requires higher-quality workers and higher wages.
- **Shock used.** The late-1994 peso crisis sharply increased the real exchange rate and made exporting to the United States much more attractive. This shock provides time-series variation in the incentive to export.
- **Cross-sectional prediction.** Within the same industry, initially larger and more productive plants should respond more strongly to the shock than smaller and less productive plants.
- **Outcomes.** The paper studies export shares, white-collar wages, blue-collar wages, the white-collar/blue-collar wage ratio, capital intensity, ISO 9000 certification, training, schooling, and other quality proxies.
- **Main evidence.** Initially larger plants increased exports, wages, wage ratios, capital intensity, and ISO 9000 certification more during 1993–1997 than initially smaller plants.
- **Comparative timing.** The same strong differential pattern is not present in adjacent non-devaluation periods, and a similar pattern appears in the earlier 1986–1989 devaluation period.
- **Economic takeaway.** Trade can raise inequality within developing-country industries, not only by changing sectoral employment composition, but by inducing high-productivity plants to upgrade quality and raise wages relative to low-productivity plants.

2 Incremental contribution

- **New mechanism for trade and inequality.** The paper contributes a within-industry quality-upgrading channel to a literature that had focused heavily on between-industry Heckscher–Ohlin reallocations, outsourcing, or skill-biased technical change.
- **Plant-level theory-to-data bridge.** It links a heterogeneous-firm model with quality differentiation directly to plant-level wage, export, and quality-proxy data.
- **Role of export destination demand.** The mechanism relies on richer Northern consumers having greater willingness to pay for quality, so exporters in a developing country choose higher quality for export markets than for domestic markets.
- **Firm heterogeneity as an inequality driver.** Rather than treating all workers in an industry alike, the paper shows how exposure to trade can generate wage dispersion across plants and within plants.
- **Use of a large exchange-rate shock.** The peso crisis supplies a quasi-experimental change in export incentives, allowing the paper to compare differential plant responses during devaluation and non-devaluation windows.
- **Multiple quality proxies.** The evidence extends beyond wages to ISO 9000 certification, training, schooling, accident/turnover/absentee outcomes, and capital intensity, strengthening the interpretation that the shock induced quality upgrading.
- **Alternative mechanisms tested.** The paper explicitly confronts alternative explanations: outsourcing/maquiladora activity, rent sharing, scale economies, productivity selection, capital-cost changes, and generic skill-biased trends.
- **Policy relevance.** The contribution reframes the winners and losers from trade in developing countries: winners are not simply unskilled workers, but workers and entrepreneurs in high-productivity export-oriented plants, especially skilled workers.

3 Data and measurement

3.1 Core datasets

- **EIA 1993–2001 panel.** The central plant panel contains 3,263 non-maquiladora manufacturing plants and measures exports, sales, employment, white-collar and blue-collar wages, capital, imports, and ownership.
- **EIA 1984–2001 balanced panel.** A longer panel of 1,114 plants allows the paper to compare the 1986–1989 devaluation, the 1989–1993 non-devaluation window, the 1993–1997 peso-crisis window, and the 1997–2001 post-crisis period.
- **EIA-ENESTyC panel.** A linked survey adds direct quality-related proxies such as ISO 9000 certification, formal training, worker schooling, turnover, absenteeism, and accident rates.
- **ENEU household survey.** Used for the aggregate wage inequality series in Figure I.
- **EMIME maquiladora panel.** Used to test whether the evidence is really driven by maquiladora outsourcing rather than quality upgrading in the non-maquiladora manufacturing sector.
- **Public-firm data.** Used to examine whether changes in the cost of capital could explain the results.

Interpretation: The key empirical strategy needs both a shock and plant-level heterogeneity. The EIA data provide plant-level export and wage changes; the peso crisis provides the shock; initial domestic sales or employment proxy for plant productivity.

3.2 Main variables

- **Initial plant productivity proxy.** Initial log domestic sales is the main proxy for the latent productivity parameter λ , because domestic sales are observed before the export response and are available across the main samples.
- **Export response.** Export share of sales and indicator-based exporting measures capture the plant's shift toward the high-quality Northern market.
- **Wage outcomes.** White-collar wages, blue-collar wages, and the white-collar/blue-collar wage ratio test whether quality upgrading raises skill-sensitive compensation.
- **Capital intensity.** K/L measures whether plants upgrade complementary inputs and technology.
- **Quality proxies.** ISO 9000 certification, formal training, schooling, turnover, accident rates, and absenteeism are used as indirect indicators of quality upgrading and workforce upgrading.

Interpretation: The paper does not observe product quality for every plant directly. Instead, it triangulates quality upgrading through export behavior, wages, capital intensity, certification, and workforce practices.

4 Model and theoretical mechanism

4.1 Demand and willingness to pay for quality

Consumers in destination $d \in \{n, s\}$ have indirect utility

$$V(\omega) = \theta_d q(\omega) - \tilde{p}_d(\omega) + \varepsilon,$$

where $q(\omega)$ is product quality, $\tilde{p}_d(\omega)$ is the local price, and θ_d is willingness to pay for quality. The core cross-country assumption is

$$\theta_n > \theta_s,$$

meaning that Northern consumers are richer and value quality more.

Interpretation: The same Mexican plant may optimally choose different quality for domestic and export markets because the export destination has stronger demand for quality.

With an extreme-value idiosyncratic taste shock, demand takes a multinomial-logit form:

$$x_d(\omega) = \frac{N_d \exp [\mu^{-1}(\theta_d q(\omega) - p_d(\omega)/\delta_d)]}{\int_{\Omega_d} \exp [\mu^{-1}(\theta_d q(\omega) - p_d(\omega)/\delta_d)] d\omega}.$$

Interpretation: Demand combines horizontal differentiation with vertical differentiation. Consumers have idiosyncratic tastes, but all else equal they prefer higher quality.

4.2 Production, worker quality, and wages

A Southern plant with productivity or ability λ produces quality

$$q_d(k_d, e_d^h, e_d^l; \lambda) = \lambda k_d^{\alpha_k} (e_d^h)^{\alpha_h} (e_d^l)^{\alpha_l},$$

where k_d is capital sophistication, e_d^h is white-collar worker quality, and e_d^l is blue-collar worker quality. Worker-quality schedules are

$$e_d^h = z_h(w_d^h - \bar{w}^h), \quad e_d^l = z_l(w_d^l - \bar{w}^l).$$

Interpretation: Higher product quality requires higher-quality labor and capital. Plants must pay higher wages to attract or motivate higher-quality workers.

4.3 Optimal plant choices

Optimization implies

$$\begin{aligned} q_d^*(\lambda) &= \eta \lambda^{1/(1-\alpha)} (\delta_d \theta_d)^{\alpha/(1-\alpha)}, \\ w_d^{h*}(\lambda) &= \bar{w}^h + \alpha_h \delta_d \theta_d q_d^*(\lambda), \\ w_d^{l*}(\lambda) &= \bar{w}^l + \alpha_l \delta_d \theta_d q_d^*(\lambda), \\ k_d^*(\lambda) &= \frac{\alpha_k}{\rho} \delta_d \theta_d q_d^*(\lambda), \quad p_d^*(\lambda) = \mu \delta_d + \bar{w}^h + \bar{w}^l + \alpha \delta_d \theta_d q_d^*(\lambda). \end{aligned}$$

Interpretation: Higher- λ plants produce higher quality, pay higher wages, use more capital-intensive production, and charge higher prices. If product quality is more sensitive to white-collar worker quality, quality upgrading raises within-plant wage inequality as well.

4.4 Export cutoffs and the peso crisis

There is a fixed cost to entering the export market, so only plants with productivity above an export cutoff export. A peso devaluation raises the peso value of export revenues and shifts export incentives upward.

$$\text{Devaluation} \Rightarrow \delta_n \uparrow \Rightarrow q_n^*(\lambda) \uparrow, \quad \lambda_n^{\min} \downarrow.$$

Interpretation: More productive plants already near or above the export cutoff have the strongest incentive to upgrade quality. New exporters and continuing exporters both experience differential quality upgrading relative to domestic-oriented plants.

5 Empirical specifications

5.1 Cross-sectional regressions

The cross-sectional specification relates initial productivity proxies to outcomes:

$$Y_{ij0} = \alpha + \beta \log(DomesticSales_{ij0}) + \Gamma_j + \Gamma_s + \varepsilon_{ij0},$$

where Γ_j and Γ_s are industry and state controls.

Interpretation: These regressions check whether initially larger plants are already higher wage, more capital intensive, more likely to use quality practices, and more likely to be export-oriented.

5.2 Differential-change regressions

The key empirical specification relates changes over a period to initial domestic sales:

$$\Delta Y_{i,t_0,t_1} = \alpha + \beta \log(DomesticSales_{i,t_0}) + \Gamma_j + \Gamma_s + u_i.$$

The main periods are 1993–1997 and 1997–2001, with a difference across periods used to isolate the peso-crisis response. The longer EIA panel also compares 1986–1989, 1989–1993, 1993–1997, and 1997–2001.

Interpretation: A positive β during the peso-crisis period means initially larger or more productive plants increased the outcome more than smaller plants. The theory predicts this for exports, wages, capital intensity, and quality proxies.

5.3 Alternative productivity proxies and robustness

The paper repeats the differential-change logic using initial log employment and other productivity-related variables, and checks whether results survive across samples, outcome definitions, and periods.

Interpretation: Because domestic sales may partly reflect demand rather than productivity, robustness to alternative proxies strengthens the interpretation that latent plant productivity drives differential responses.

6 Identification and empirical design

6.1 Core identification logic

- The peso crisis generated a large and plausibly external change in relative export incentives for Mexican manufacturers.
- The key comparison is within industries and states: initially large plants versus initially small plants facing the same broad macro shock.
- The theory predicts that productivity interacts with the shock. A uniform devaluation should have heterogeneous effects by initial productivity because only high- λ plants can profitably upgrade and export.
- Adjacent non-devaluation periods are used as placebo or contrast windows. If the result were just a general trend, it should appear equally in 1989–1993 and 1997–2001.

Interpretation: The strategy is a differential exposure design: the same aggregate shock hits all plants, but the model predicts stronger treatment intensity for initially more productive plants.

6.2 Main threats and responses

- **Selection and exit.** If low-productivity plants exit, changes among survivors may be mechanically different. The paper checks robustness with selection corrections and alternative samples.
- **Rent sharing.** If large exporters simply earn higher profits and share rents with workers, wages may rise without quality upgrading. The paper uses quality proxies such as ISO 9000 and training to support the quality interpretation.
- **Scale economies.** If larger plants benefit more from scale, wages might rise mechanically. The table patterns and quality proxies help distinguish this from the export-quality mechanism.
- **Outsourcing/maquiladora channel.** The Feenstra–Hanson outsourcing mechanism should be strongest in maquiladoras, but the paper finds the relevant differential wage pattern in non-maquiladora plants, not in maquiladoras during the peso-crisis period.
- **Capital-cost channel.** If larger plants face different financing shocks, capital-cost changes might drive results. The public-firm tests provide little evidence of such a channel.

Interpretation: The evidence is strongest because many independent outcomes move in the direction predicted by quality upgrading: exporting, wages, wage ratios, capital intensity, ISO certification, and worker-quality practices.

7 Section-by-section study guide

- **Introduction.** States the puzzle: trade expanded and wage inequality rose, despite the simplest trade theory predicting the opposite for Mexico.
- **Peso crisis background.** Establishes the 1994 devaluation as a large change in export incentives and explains why it is more useful empirically than NAFTA tariff phase-ins alone.
- **Theory.** Shows how heterogeneous plants, quality-differentiated goods, and wage-quality schedules imply that high-productivity exporters upgrade quality and wages more after devaluation.
- **Data.** Introduces plant panels and quality proxies, emphasizing that maquiladoras are excluded from the main non-maquiladora EIA sample.
- **Empirical results.** Tests whether initially larger plants differentially increase exports, wages, wage ratios, capital intensity, and quality proxies during devaluation periods.
- **Alternative hypotheses.** Uses cost-of-capital and maquiladora tests to rule out competing mechanisms.
- **Conclusion.** Recasts trade winners and losers as workers and entrepreneurs in modern export-oriented plants versus those in lower-productivity domestic plants.

8 Tables and figures

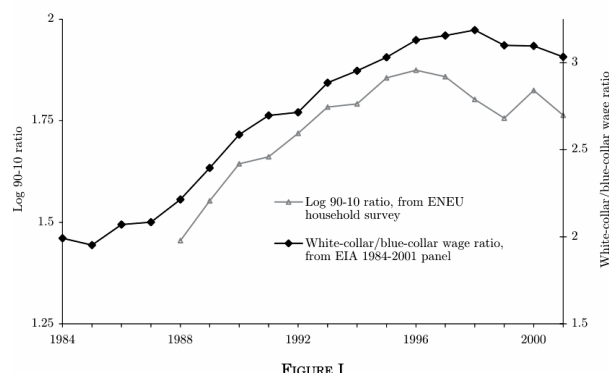
8.1 Figure I: Wage Inequality, 1984–2001 + Figure II: Wage Variance, EIA 1984–2001 Panel

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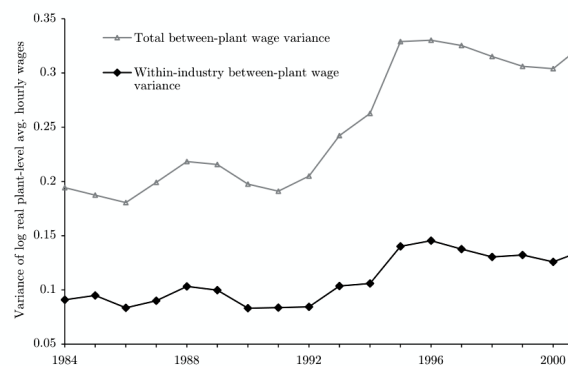
- Figure I shows the log 90–10 wage ratio and the white-collar/blue-collar wage ratio rising sharply from the late 1980s to the mid-1990s.
- The peak occurs around 1996–1998, immediately after the peso crisis and export surge.
- Figure II shows that plant-level wage dispersion also rises, both in total and within industries.
- The within-industry component matters because the paper’s mechanism operates across plants inside the same industry.

Detailed interpretation: The two figures establish the dependent-variable puzzle. Wage inequality is not only rising in the economy as a whole; wage dispersion across manufacturing plants is also rising. This motivates a mechanism based on plant heterogeneity, not merely a shift from one industry to another.

Economic message: The paper’s core puzzle is within-industry inequality: trade shocks appear to separate high-productivity, high-quality plants from lower-productivity domestic plants.



(a) Figure I: Wage Inequality, 1984–2001



(b) Figure II: Wage Variance, EIA 1984–2001 Panel

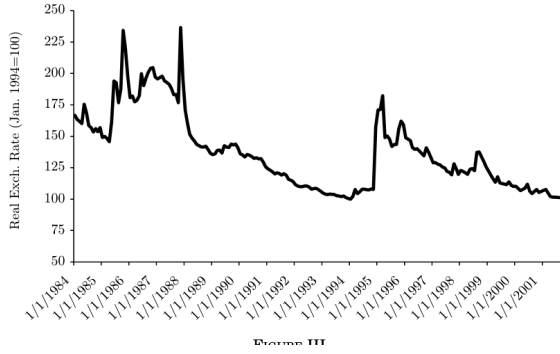
8.2 Figure III: Real Exchange Rate, 1984–2002 + Figure IV: Exports, High-quality Models as Percentage of VW Output

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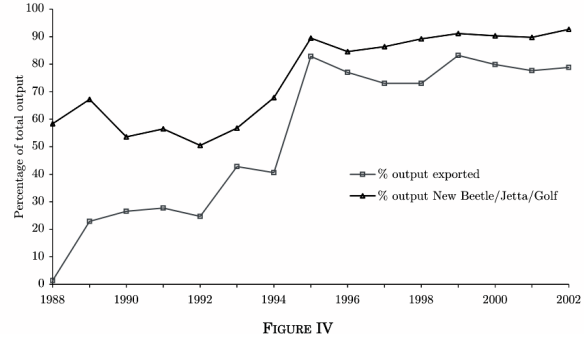
- Figure III shows a major real peso depreciation in 1994–1995, much larger and more abrupt than NAFTA’s gradual tariff changes.
- Figure IV uses the Volkswagen Puebla plant as a concrete example: exports and high-quality models jump strongly after the peso crisis.
- The VW example illustrates the paper’s mechanism in micro form: exporting to richer consumers goes together with producing higher-quality models.

Detailed interpretation: The exchange-rate figure identifies the macro shock. The VW figure provides a case-study bridge from the macro shock to the plant-level mechanism. Together, they support the idea that the devaluation changed the return to export-oriented quality upgrading.

Economic message: The 1994 peso crisis is not just a financial event; it changes relative prices in a way that makes high-quality exporting more profitable for Mexican plants.



(a) Figure III: Real Exchange Rate, 1984–2002



(b) Figure IV: Exports, High-quality Models as Percentage of VW Output

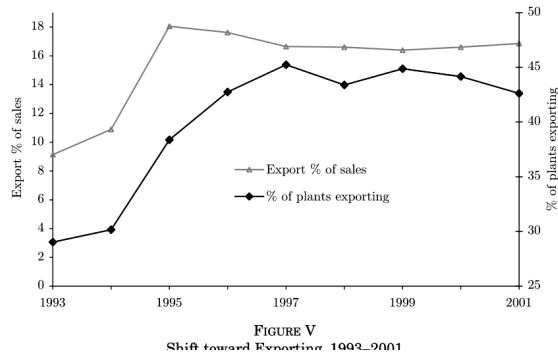
8.3 Figure V: Shift toward Exporting, 1993–2001 + Figure VI: Average Quality

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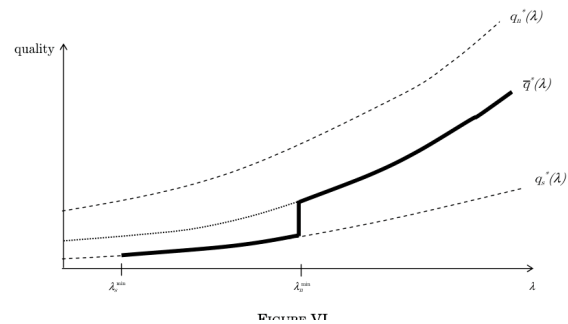
- Figure V shows that both the export share of sales and the share of plants exporting rise sharply from 1993 to 1995 and remain high afterward.
- Figure VI translates the model into a cross-sectional prediction: average observed quality rises with plant productivity, and export entry creates a discrete jump in observed average quality.
- The solid curve in Figure VI is the empirical object: observed plant average quality, not separate production-line quality.

Detailed interpretation: Figure V documents the aggregate shift toward exporting that the empirical regressions exploit. Figure VI shows why the cross-section matters: high-productivity plants have higher quality even before the shock, and the export cutoff produces a nonlinearity. This explains why initial domestic sales can proxy for latent productivity.

Economic message: The export boom is the shock; the productivity-quality schedule is the mechanism translating the shock into wage inequality.



(a) Figure V: Shift toward Exporting, 1993–2001



(b) Figure VI: Average Quality

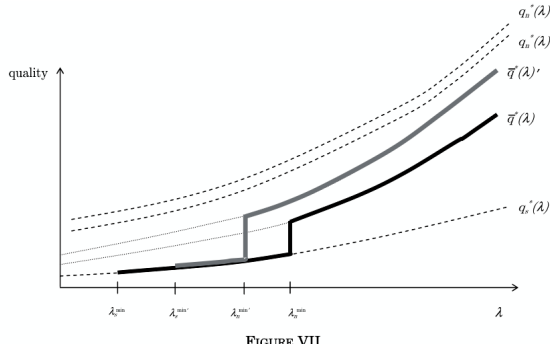
8.4 Figure VII: Response to Exchange-Rate Devaluation + Figure VIII: Change in Average Quality in Response to Devaluation

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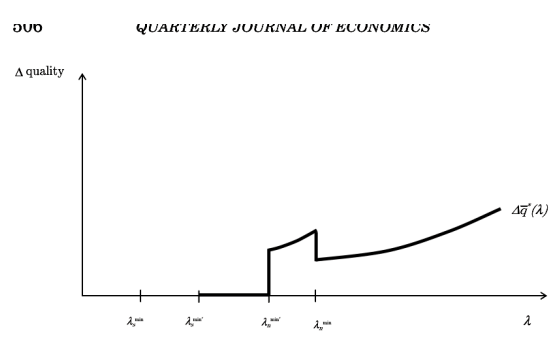
- Figure VII shows the theoretical effect of devaluation: export quality rises and the export cutoff shifts left, so more plants export.
- Figure VIII shows the resulting change in average observed quality by plant productivity.
- Plants just induced to export experience a large discrete quality jump; continuing exporters also upgrade gradually with productivity.
- Low-productivity plants that remain domestic should show little or no quality change.

Detailed interpretation: These two figures are the theoretical heart of the paper. The empirical tables are essentially tests of the Figure VIII shape using exports, wages, wage ratios, capital intensity, and quality proxies as observed outcomes.

Economic message: The mechanism predicts a differential response, not merely an economy-wide wage change: the higher the initial productivity, the stronger the quality-upgrading response.



(a) Figure VII: Response to Exchange-Rate Devaluation



(b) Figure VIII: Change in Average Quality in Response to Devaluation

8.5 Table I: Summary Statistics by Export Status + Table II: Baseline Estimates, EIA 1993–2001 Panel

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- Table I shows exporters are larger than nonexporters: exporters have higher employment, revenues, domestic sales, capital intensity, and wages.
- Exporters also pay higher white-collar and blue-collar wages and have a higher import share of input expenditures and foreign ownership.
- Table II shows that initial domestic sales strongly predicts initial wages, wage ratios, capital intensity, and white-collar employment share.
- During 1993–1997, initially larger plants raise export share, white-collar wages, blue-collar wages, wage ratios, and capital intensity more than smaller plants.
- The 1997–2001 coefficients are generally smaller, and differences between periods support the peso-crisis interpretation.

Detailed interpretation: Table I validates the exporter/nonexporter distinction and the productivity proxy. Table II is the paper's central empirical table: it maps the model's λ predictions into plant-level regressions. The strongest part of the table is that export shares and both wage categories rise differentially for initially large plants exactly during the peso-crisis window.

Economic message: Exporting plants were already different, but the crisis induced them to diverge further from domestic plants.

TABLE I
SUMMARY STATISTICS BY EXPORT STATUS

	Nonexporters	Exporters	All
A. EIA 1993–2001 Panel, 1993			
Employment	182.39 [4.94]	333.76 [11.98]	226.32 [5.08]
Revenues	42.24 [1.75]	89.84 [4.59]	56.05 [1.86]
Domestic sales	41.17 [1.71]	70.16 [3.59]	49.58 [1.61]
K/L ratio	42.58 [1.40]	55.59 [2.60]	46.36 [1.25]
White-collar hourly wage	20.53 [0.27]	28.40 [0.50]	22.81 [0.25]
Blue-collar hourly wage	8.04 [0.08]	9.64 [0.15]	8.50 [0.07]
White-collar employment share	0.30 [0.004]	0.33 [0.005]	0.31 [0.003]
Export share of sales		0.16 [0.01]	
Import share of input expenditures	0.14 [0.01]	0.30 [0.01]	0.19 [0.005]
Share with foreign ownership	0.09 [0.01]	0.30 [0.01]	0.15 [0.01]
<i>N</i>	2,316	947	3,263
B. EIA-ENESTyC Panel			
Employment	308.37 [12.18]	414.29 [21.08]	348.91 [11.16]
Share with ISO 9000 certification	0.09 [0.01]	0.12 [0.02]	0.10 [0.01]
Share with formal training program	0.69 [0.02]	0.79 [0.02]	0.73 [0.02]
Avg. schooling, white-collar	12.04 [0.08]	12.47 [0.10]	12.20 [0.06]
Avg. schooling, blue-collar	7.26 [0.07]	7.38 [0.09]	7.30 [0.05]
Absentee rate	1.41 [0.07]	1.20 [0.08]	1.33 [0.05]
Turnover rate	74.70 [3.66]	66.62 [4.44]	71.56 [2.83]
Accident rate	5.01 [0.29]	4.63 [0.31]	4.87 [0.21]

Notes. Standard errors of means in brackets. Exporter means export sales > 0. Has foreign ownership means FDI share > 0. Data on FDI from 1994; other data in Panel A from EIA 1993–2001 panel for 1993. Revenues, domestic sales measured in millions of 1994 pesos, K/L ratio in thousands of 1994 pesos, wages in 1994 pesos per hour, employment in number of workers, employment share in hours. Average 1994 exchange

Table I: Summary Statistics by Export Status

8.6 Table III: Estimates from EIA 1984–2001 Panel + Table IV: Estimates from EIA-ENESTyC Panel

Read:

- Table III extends the baseline logic to four periods, including the earlier 1986–1989 devaluation.
- The 1986–1989 and 1993–1997 devaluation windows show stronger differential wage and export responses than the non-devaluation windows.
- Table IV uses quality proxies: ISO 9000 certification, worker schooling, formal training, turnover, accident rates, and absenteeism.
- Initially larger plants are more likely to be certified, have more schooling, offer more training, and have better workplace outcomes.
- During 1993–1997, initially larger plants increase ISO certification substantially more than smaller plants.

Detailed interpretation: Table III addresses timing: the same kind of differential response appears in devaluation periods and is weaker in non-devaluation periods. Table IV addresses mechanism: ISO certification is a direct sign of quality-oriented organizational upgrading, while training and schooling capture workforce upgrading.

TABLE II BASELINE ESTIMATES, EIA 1993–2001 PANEL						
A. Cross-sectional regressions, 1993						
	Export share of sales (1)	log(white- collar wage) (2)	log(blue- collar wage) (3)	log(wage ratio) (4)	log(K/L ratio) (5)	White-collar emp. share (6)
log domestic sales, 1993	-0.001 (0.003)	0.209*** (0.006)	0.132*** (0.004)	0.073*** (0.004)	0.343*** (0.012)	0.010*** (0.002)
<i>R</i> ²	0.230	0.391	0.358	0.348	0.370	0.343
B. Differential changes, 1993–1997 and 1997–2001						
	Δ (export share of sales) (1)	Δ log(white- collar wage) (2)	Δ log(blue- collar wage) (3)	Δ log(wage ratio) (4)	Δ log(K/L ratio) (5)	Δ (white-collar emp. share) (6)
OLS regressions						
1993–1997	log domestic sales, 1993	0.020*** (0.002)	0.072*** (0.006)	0.036*** (0.006)	0.053*** (0.011)	-0.002 (0.002)
<i>R</i> ²		0.173	0.15	0.129	0.09	0.134
1997–2001	log domestic sales, 1997	0.067*** (0.002)	0.034*** (0.007)	0.008 (0.005)	0.048 (0.007)	0.038*** (0.001)
<i>R</i> ²		0.123	0.088	0.092	0.075	0.107
Differences (1993–1997 vs. 1997–2001)		0.044*** (0.003)	0.036*** (0.016)	0.028*** (0.007)	0.028*** (0.011)	0.067*** (0.014)
IV regressions						
1993–1997	log domestic sales, 1994	0.014*** (0.003)	0.058*** (0.007)	0.032*** (0.003)	0.038*** (0.008)	0.008 (0.003)
<i>R</i> ²		0.181	0.148	0.118	0.093	0.119
1997–2001	log domestic sales, 1998	0.064*** (0.002)	0.035*** (0.006)	0.004 (0.004)	0.050*** (0.007)	0.038*** (0.001)
<i>R</i> ²		0.111	0.082	0.097	0.077	0.102
Differences (1993–1997 vs. 1997–2001)		0.049*** (0.003)	0.023*** (0.006)	0.029*** (0.007)	0.024*** (0.010)	0.042*** (0.012)

Notes. Table reports coefficients on log domestic sales for 30 separate regressions. Covariates at left, dependent variables at top, with changes in Panel B over period at left. All regressions include 500 industry (six-digit) and 52 state dummies. IV regressions instrument log domestic sales in 1994 and 1998 with values from previous year. *N* = 3,263 for all regressions. Variable definitions in Appendix I. Further details on data set in Section IV of the text and Appendix II (online). Robust standard errors in brackets. Standard errors on differences allow for cross-equation correlation. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table II: Baseline Estimates, EIA 1993–2001 Panel

Economic message: The quality-upgrading interpretation is stronger because the evidence is not only about wages; it also appears in certification and workforce-quality measures.

TABLE III
ESTIMATES FROM EIA 1984–2001 PANEL

		Δ (export share of sales) (1)	Δ log(white-collar wage) (2)	Δ log(blue-collar wage) (3)	Δ log(wage ratio) (4)	Δ log(KUL ratio) (5)	Δ (white-collar emp. share) (6)
1986–1989	log domestic sales, 1986	0.011*** [0.002]	0.047*** [0.010]	0.020*** [0.008]	0.023** [0.011]	0.040*** [0.015]	-0.006** [0.003]
	R^2	0.185	0.182	0.254	0.194	0.208	0.207
1989–1993	log domestic sales, 1989	0.007*** [0.003]	0.004 [0.011]	0.006 [0.009]	0.002 [0.013]	0.076*** [0.027]	-0.001 [0.003]
	R^2	0.219	0.173	0.211	0.167	0.179	0.195
1993–1997	log domestic sales, 1993	0.013*** [0.005]	0.067*** [0.012]	0.021** [0.008]	0.045*** [0.012]	0.092*** [0.017]	0.002 [0.003]
	R^2	0.277	0.272	0.236	0.188	0.266	0.219
1997–2001	log domestic sales, 1997	0.007*** [0.003]	0.002 [0.009]	0.009 [0.008]	-0.009 [0.011]	0.014 [0.014]	-0.001 [0.002]
	R^2	0.200	0.193	0.217	0.163	0.188	0.146

Notes: Table reports coefficients on log domestic sales for 24 separate regressions. Coefficient at left, dependent variable at top, with changes over period at left. All regressions include 303 industry (two-digit) and 32 state domestic coefficients omitted. $N = 1,114$ for all regressions. Variable definitions in Appendix I. Further details on data set in Section IV of the text and Appendix II (online). Robust standard errors in brackets. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table III: Estimates from EIA 1984–2001 Panel

TABLE IV
ESTIMATES FROM EIA-ENESTyC PANEL

A. Cross-sectional regressions, 1993							
	ISO 9000 certification (1)	White-collar avg. schooling (2)	Blue-collar avg. schooling (3)	Has formal training (4)	Turnover rate (5)	Accident rate (6)	Absentee rate (7)
Log domestic sales, 1993	0.023** [0.011]	0.286*** [0.067]	0.156*** [0.056]	0.049*** [0.017]	-20.239*** [2.965]	-0.802*** [0.216]	-0.250*** [0.044]
N	844	590	843	751	828	515	828
R^2	0.154	0.258	0.240	0.117	0.168	0.206	0.245
B. Differential Responses, 1993–1997 and 1997–2001							
	Δ ISO 9000 certification (1)	Δ white-collar avg. schooling (2)	Δ blue-collar avg. schooling (3)	Δ has formal training (4)	Δ turnover rate (5)	Δ accident rate (6)	Δ absentee rate (7)
1993–1997 Log domestic sales, 1993	0.079*** [0.018]	-0.105 [0.018]	0.204*** [0.078]	0.008 [0.020]	1.087 [4.224]	0.219 [0.247]	-0.025 [0.093]
	R^2	0.171	0.164	0.194	0.1	0.184	0.141
1997–2001 Log domestic sales, 1997	0.036*** [0.015]	0.058 [0.015]	-0.023 [0.075]	-0.024 [0.017]	-4.294 [4.655]	0.045 [0.222]	-0.140 [0.093]
	R^2	0.127	0.151	0.173	0.082	0.161	0.134
Difference (1993–1997 vs. 1997–2001)	0.042* [0.024]	-0.163 [0.138]	0.228** [0.109]	0.032 [0.026]	5.361 [8.286]	0.174 [0.232]	0.115 [0.131]
N	844	484	484	536	513	713	354

Notes: Table reports coefficients on log domestic sales for 24 separate regressions. Coefficient at left, dependent variable at top, with changes in Panel B over period at left. All regressions include dummies for 50 industries (two-digit) and 32 states. Data on ISO 9000, training, turnover rate, accident rate, absentee rate from 1993, 1996, 1999, 2000. Since reporting plants to have complete data on all variables would have reduced the panel substantially, I follow the sample size to change across columns. Variable definitions in Appendix I. Further details on data set in Section IV of the text and Appendix II (online). Robust standard errors in brackets. Standard errors on difference allow for cross-equation correlation. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table IV: Estimates from EIA-ENESTyC Panel

8.7 Table V: Estimates Using Alternative Proxies for λ , EIA 1993–2001 Panel + Table VI: Estimates for Cost of Capital, Publicly Listed Mexican Firms, 1989–2000

Read:

- Table V checks whether the main results depend on using initial domestic sales as the productivity proxy.
- Alternative proxies such as employment, revenues, and value-added measures deliver similar positive differential responses during 1993–1997.
- Table VI examines whether changes in the cost of capital can explain the patterns.
- The cost-of-capital tests do not show a pattern consistent with larger plants gaining systematically better financing during the crisis.

Detailed interpretation: Table V strengthens the productivity interpretation because the results are not an artifact of a single proxy. Table VI addresses a competing mechanism: if larger firms had lower cost of capital after the shock, their capital and wage growth might not reflect quality upgrading. The evidence does not favor that explanation.

Economic message: Robustness to proxy choice and rejection of a financing-cost explanation support the quality-upgrading channel.

8.8 Table VII: Comparing Non-maquiladora and Maquiladora Sectors

Read:

- Table VII compares the main EIA non-maquiladora sector with the maquiladora sector using the EMIME panel.
- In the non-maquiladora sector, initially larger plants have stronger wage growth during 1993–1997 than during 1997–2001.
- In the maquiladora sector, the pattern is reversed or absent during the peso-crisis period.
- This weakens the interpretation that the main results are driven by the Feenstra–Hanson outsourcing mechanism.

TABLE V
ESTIMATES USING ALTERNATIVE PROXIES, EIA 1993-2001 PANEL

	Δ (import share of sales)			Δ log(white-collar wage)			Δ log(blue-collar wage)			Δ log(wage ratio)		
	1993-1997	1997-2001	Difference	1993-1997	1997-2001	Difference	1993-1997	1997-2001	Difference	1993-1997	1997-2001	Difference
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Proxy for λ :												
Log employment	0.017***	0.004*	0.013***	0.088***	0.013	0.075***	0.050***	0.018***	0.033***	0.030***	-0.005	0.040***
	[0.003]	[0.002]	[0.004]	[0.010]	[0.006]	[0.014]	[0.007]	[0.004]	[0.010]	[0.012]	[0.006]	[0.010]
R^2	0.156	0.12	0.147	0.087	0.131	0.093	0.131	0.088	0.075	0.088	0.075	0.088
Predicted exp. share index	0.017***	0.003	0.014***	0.088***	0.013**	0.080***	0.048***	0.013***	0.033***	0.023***	0.000	0.031***
	[0.003]	[0.002]	[0.003]	[0.009]	[0.007]	[0.011]	[0.006]	[0.005]	[0.008]	[0.006]	[0.006]	[0.010]
R^2	0.162	0.119	0.154	0.087	0.132	0.093	0.137	0.093	0.088	0.075	0.088	0.075
Predicted ISO 9000 index	0.006***	-0.005**	0.011***	0.088***	0.011	0.078***	0.024***	0.003	0.021**	0.007***	0.008	0.023***
	[0.003]	[0.002]	[0.003]	[0.009]	[0.006]	[0.012]	[0.007]	[0.006]	[0.006]	[0.010]	[0.008]	[0.013]
R^2	0.143	0.121	0.151	0.087	0.132	0.091	0.122	0.091	0.094	0.075	0.094	0.075
First principal component	0.013***	-0.001	0.013***	0.084***	0.020***	0.064***	0.044***	0.010*	0.034***	0.020***	0.009	0.030***
	[0.003]	[0.002]	[0.003]	[0.009]	[0.007]	[0.011]	[0.006]	[0.005]	[0.008]	[0.006]	[0.006]	[0.010]
R^2	0.156	0.119	0.150	0.088	0.134	0.092	0.134	0.092	0.09	0.075	0.09	0.075
TVP (fixed effects)	0.013***	0.002	0.011***	0.060***	0.013*	0.047***	0.020***	0.002	0.027***	0.033***	0.01	0.021
	[0.003]	[0.002]	[0.003]	[0.009]	[0.006]	[0.012]	[0.006]	[0.006]	[0.009]	[0.010]	[0.009]	[0.010]
R^2	0.156	0.119	0.144	0.087	0.135	0.091	0.135	0.091	0.088	0.075	0.088	0.075
Log (dom sales/worker)	0.020***	0.012***	0.017***	0.050***	0.022**	0.038***	0.017*	-0.006	0.023*	0.043***	0.027**	0.013
	[0.004]	[0.002]	[0.003]	[0.010]	[0.011]	[0.017]	[0.010]	[0.009]	[0.010]	[0.010]	[0.010]	[0.010]
R^2	0.166	0.125	0.154	0.088	0.139	0.091	0.139	0.091	0.087	0.076	0.087	0.076
Export share	0.144**	0.029	0.115	0.173***	0.050	0.173***	-0.028	0.023	-0.06			
	[0.007]	[0.048]	[0.076]	[0.023]	[0.040]	[0.050]	[0.061]	[0.051]	[0.051]			
R^2	0.120	0.096	0.121	0.091	0.121	0.091	0.085	0.075				

Notes: Table reports coefficients on log domestic sales for 12 separate regressions. Covariate at left, dependent variable at top, with changes over period at top. All regressions include 1993 industry (six-digit) and 22 state dummies. Coefficients when initial export share is proxy and change in export share is outcome omitted because measurement error generates severe biases. Variable definitions in Appendix I. Further details on data sets in Section IV of the text and Appendix II (online). Robust standard errors in brackets. Standard errors on differences allow for cross-equation correlation. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table V: Estimates Using Alternative Proxies for λ

TABLE VI
ESTIMATES FOR COST OF CAPITAL, PUBLICLY LISTED MEXICAN FIRMS, 1989-2000

		Dep. var.: Δ cost of capital			
		All sectors		Manufacturing only	
		OLS	IV	OLS	IV
		(1)	(2)	(3)	(4)
1989-1993	Log domestic sales, 1989	1.545	-1.13	-1.483	-1.724
		[1.850]	[1.170]	[2.353]	[1.649]
	N	68	68	37	37
	R^2	0.139	0.042	0.011	0.005
1993-1997	Log domestic sales, 1993	0.22	0.379	0.088	0.397
		[0.671]	[0.414]	[1.059]	[0.554]
	N	122	122	57	57
	R^2	0.086	0.128	0.001	0.010
1997-2000	Log domestic sales, 1997	0.642	1.473	0.155	0.948*
		[0.411]	[0.981]	[0.471]	[0.510]
	N	113	113	52	52
	R^2	0.050	0.083	0.002	0.088

Notes: Table reports coefficients on initial log domestic sales for 12 separate regressions. Covariate at left, dependent variable at top, with changes over period at top. Regressions for all sectors together include industry effects (six coarse industry categories). IV regression instrument log domestic sales in 1990, 1994, and 1998 with values from previous year, second-stage dependent variables are changes over 1989-1993, 1994-1997, and 1998-2001, respectively. Cost of capital defined as $100 \times$ interest paid/total debt. Data set is unbalanced panel of publicly listed firms on Mexican stock market (Bolsa Mexicana de Valores) for 1989-2000. For further details on data set, see Prasad and Uralina (2004) and Aguilar (2005). Robust standard errors in brackets. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table VI: Estimates for Cost of Capital

TABLE VII
COMPARING NON-MAQUILADORA AND MAQUILADORA SECTORS

		EIA 1993-2001 panel (non-maquiladora sector)			EMIME panel (maquiladora sector)		
		Δ log(average hourly wage)	Δ log(white-collar yearly earnings)	Δ log(blue-collar yearly earnings)	Δ log(average hourly wage)	Δ log(white-collar yearly earnings)	Δ log(blue-collar yearly earnings)
		(1)	(2)	(3)	(4)	(5)	(6)
1993-1997	Log employment 1993	0.072***	0.055***	0.050***	-0.019	-0.023	-0.029**
		[0.007]	[0.010]	[0.007]	[0.016]	[0.016]	[0.014]
	R^2	0.157	0.124	0.140	0.055	0.044	0.041
1997-2001	Log employment 1997	0.025***	0.013	0.017***	0.032**	0.050***	0.038***
		[0.006]	[0.008]	[0.006]	[0.014]	[0.014]	[0.013]
	R^2	0.109	0.086	0.104	0.047	0.049	0.061
Difference (1993-1997 vs. 1997-2001)		0.048***	0.042***	0.033***	-0.050**	-0.073***	-0.067***
		[0.009]	[0.013]	[0.009]	[0.021]	[0.021]	[0.019]
N		3,263	3,263	3,263	1,088	1,088	1,088

Notes: Table reports coefficients on initial log employment for 12 separate regressions. Covariate at left, dependent variables at top, with changes over period at top. Columns (1)-(3) use the EIA 1993-2001 panel and include 205 industry (six-digit) and 32 state dummies. Columns (4)-(6) use the EMIME panel over same period and include 12 industry and 32 state dummies. Variable definitions in Appendix I. Further details on data sets in Section IV of the text and Appendix II (online). Robust standard errors in brackets. Standard errors on differences allow for cross-equation correlation. *** indicates significance at 1% level, ** at 5% level, * at 10% level.

Table VII: Comparing Non-maquiladora and Maquiladora Sectors

Detailed interpretation: If outsourcing were the dominant mechanism, the maquiladora sector should show the strongest differential skill/wage response because maquiladoras specialize in export processing for Northern firms. Instead, the main differential pattern is in the non-maquiladora sector, where quality upgrading by exporters is more plausible.

Economic message: The main results are not simply about maquiladora outsourcing; they are about export-induced quality upgrading within Mexico's broader manufacturing sector.

9 Result map

- **Aggregate wage inequality rises.** Figure I shows economy-wide and manufacturing-sector wage inequality increasing into the mid-1990s.
- **Within-industry plant wage dispersion rises.** Figure II shows that the dispersion of plant-level wages rises even after industry-year effects are removed.
- **The peso crisis is the key shock.** Figure III identifies a sharp, discrete devaluation that changed export incentives.

- **Exporting shifts sharply.** Figure V shows a large increase in export share and exporter participation after 1994.
- **Model prediction.** Figures VI–VIII imply that high-productivity plants should upgrade quality and wages more strongly after devaluation.
- **Baseline evidence.** Table II confirms that initially larger plants increased exports, wages, wage ratios, and capital intensity more during 1993–1997.
- **Timing evidence.** Table III shows similar patterns in the earlier devaluation period and weaker patterns in non-devaluation windows.
- **Quality-proxy evidence.** Table IV shows differential increases in ISO certification and other workforce-quality measures.
- **Robustness.** Table V shows results do not depend on a single productivity proxy.
- **Alternative hypotheses.** Tables VI and VII weaken cost-of-capital and outsourcing interpretations.

10 Short summary

- The paper explains why trade expansion can raise wage inequality in a developing country.
- It introduces a quality-upgrading mechanism: more productive exporters make higher-quality goods for richer foreign consumers.
- Higher-quality production requires higher-quality workers and higher wages.
- The peso crisis increased export incentives and caused more productive Mexican plants to shift toward exporting.
- Initially larger plants increased exports, wages, wage ratios, capital intensity, and ISO 9000 certification more than smaller plants during 1993–1997.
- The pattern is stronger in devaluation periods than in non-devaluation periods.
- The results survive alternative productivity proxies and are not explained well by cost-of-capital or maquiladora outsourcing mechanisms.
- The key policy implication is that globalization may benefit skilled workers in modern export-oriented plants while leaving workers in less productive domestic plants behind.

11 Conclusion

11.1 Core conclusion

Verhoogen argues that trade can increase wage inequality in developing countries through within-industry quality upgrading. When export opportunities expand, more productive plants shift toward higher-quality production for richer foreign consumers, upgrade capital and labor inputs, and raise wages relative to less productive plants.

11.2 Economic intuition

The mechanism is intuitive: foreign consumers, especially in the United States, value quality more than domestic Mexican consumers. When the peso devaluation makes exporting more profitable, productive Mexican plants find it worthwhile to produce higher-quality goods. Because producing these goods requires better capital and higher-quality workers, the plants pay higher wages. Since productive plants already pay higher wages, the shock widens wage dispersion across plants and can also increase wage gaps within plants.

11.3 Incremental contribution relative to standard trade theory

The paper moves beyond the prediction that trade affects inequality mainly through reallocations across industries. Its central contribution is a within-industry, firm-heterogeneity channel. Exporting does not merely shift employment toward a sector; it changes which plants within an industry upgrade quality and who within those plants benefits.

11.4 Evidence assessment

The empirical case is persuasive because the paper combines a large macro shock, plant-level panel data, multiple outcome variables, multiple periods, and multiple quality proxies. The same differential pattern appears in exports, wages, wage ratios, capital intensity, and ISO certification. The timing patterns also line up with devaluation windows.

11.5 Limitations

The paper cannot observe product quality for every plant-year directly and must rely on proxies such as ISO certification, wages, capital intensity, and training. Initial domestic sales is an imperfect proxy for productivity. The devaluation also came with a recession and many macro changes, so the design depends on comparing differential responses by initial productivity within industries and across periods.

11.6 Takeaway

The lasting lesson is that trade liberalization can create new inequality within developing-country manufacturing by rewarding workers and entrepreneurs in modern export-oriented plants more than workers in domestic-oriented plants. Globalization's winners and losers are therefore defined not only by skill or sector, but by plant productivity, export access, and quality upgrading.